Unit DoF6

SKU:U228























Description

Unit DoF6 is a six-axis inertial sensor unit built around the low-power BMI270 IMU. The BMI270 integrates a 16-bit 3-axis accelerometer and a 16-bit 3-axis gyroscope, with configurable measurement ranges, on-chip motion detection, and fast startup. Data is transmitted over I2C, enabling real-time acquisition of acceleration, angular velocity, and temperature for motion-sensing applications such as robot attitude sensing, motion detection, wearables, gimbal control, and human-machine interaction.

Features

- BMI270 six-axis inertial sensor
- I2C interface with address switching support
- 16-bit accelerometer and gyroscope sampling
- Configurable measurement ranges and built-in motion detection
- Acceleration, angular velocity, and temperature acquisition
- Suited to attitude sensing and motion detection

Includes

- 1 x Unit DoF6
- 1 x HY2.0-4P Grove Cable (5cm)

Applications

- Robot attitude sensing
- Motion detection
- Wearables
- Gimbal control
- Human-machine interaction

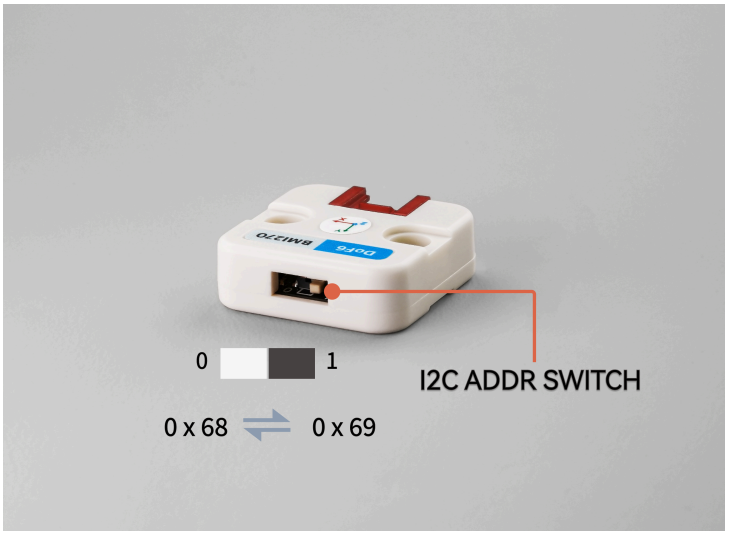
Specifications

Specification	Parameter
IC Model	BMI270
Sensor Type	Six-axis inertial sensor
Accelerometer	16-bit 3-axis accelerometer
Accelerometer Range	±2 g / ±4 g / ±8 g / ±16 g
Accelerometer Accuracy	±20 mg (approx. ±0.02 g)
Gyroscope	16-bit 3-axis gyroscope
Gyroscope Range	±125 / ±250 / ±500 / ±1000 / ±2000 °/s
Gyroscope Accuracy	±0.5 °/s
Communication Interface	I2C @ 0x68 / 0x69 (switchable via onboard switch)
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.1g
Package Size	138.0 x 93.0 x 7.0mm
Gross Weight	6.6g

Learn

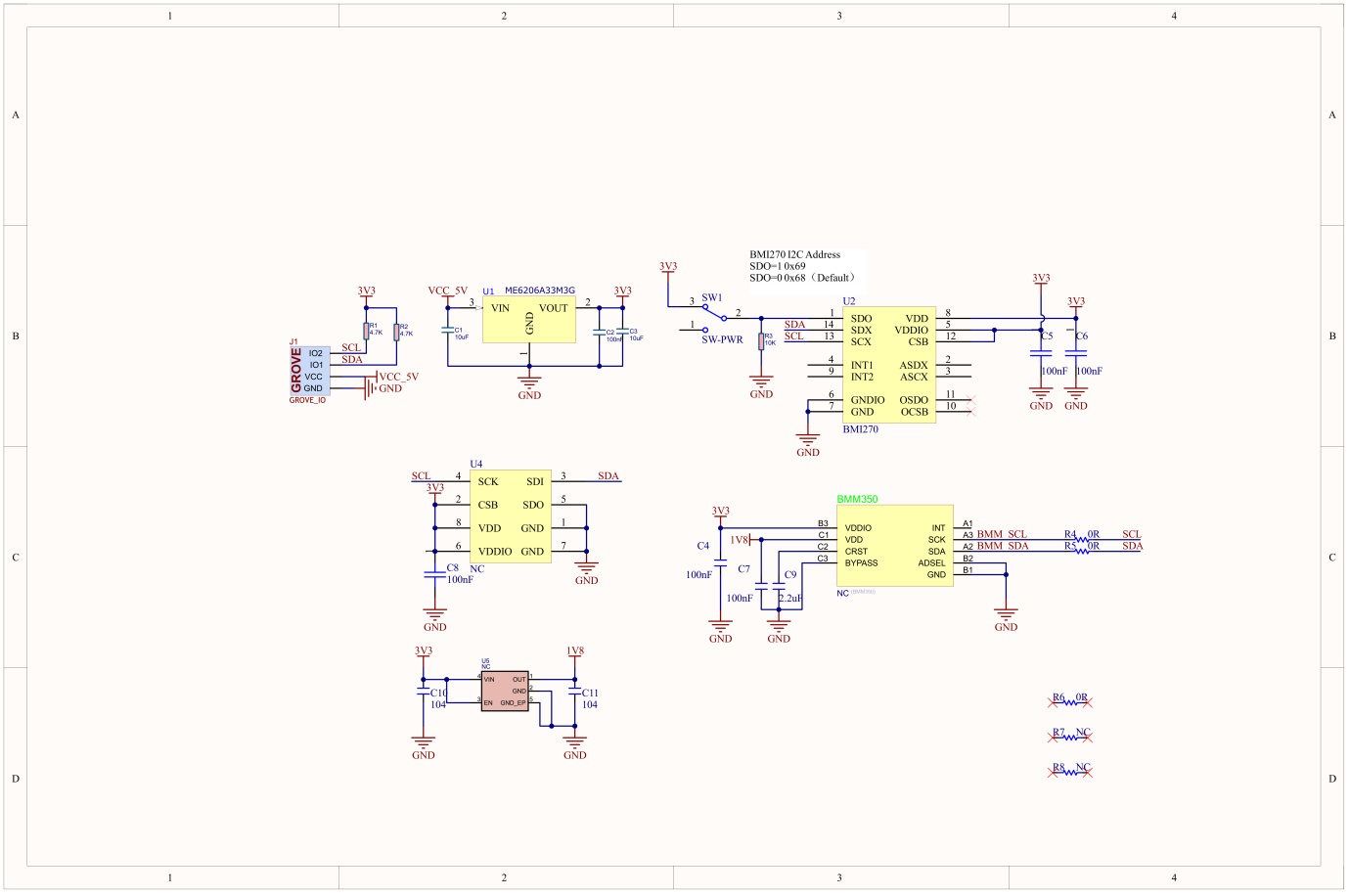
I2C Address Switching

The BMI270 I2C address can be switched with the onboard switch. Set the switch to `0` for address `0x68`, or to `1` for address `0x69`, to avoid address conflicts on the I2C bus.



Schematics

- Unit DoF6 Schematics PDF



PinMap

Unit DoF6

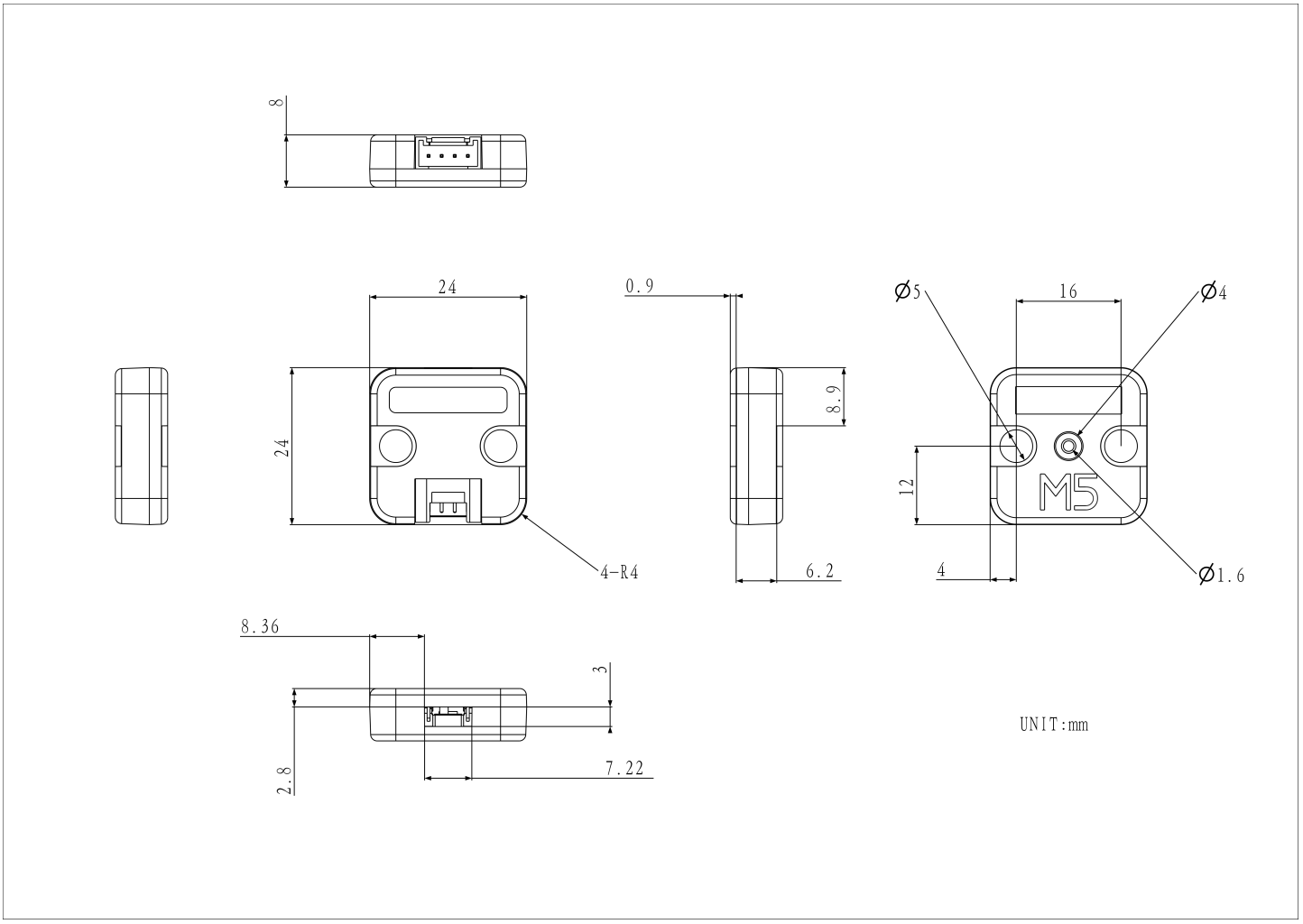
HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Datasheets

- BMI270 Datasheet

Model Size

- Unit DoF6 Model Size PDF



Structure

- Unit DoF6 Structure Files

Softwares

Arduino

- Unit DoF6 Arduino Quick Start
- Unit DoF6 Arduino Library
- M5Unified
- M5UnitUnified

UiFlow2

- Unit DoF6 UiFlow2 Docs

Video

Product Comparison

Product Comparison				
	Unit DoF6	**Unit DoF9**	**Unit DoF10**	**Unit BMM350**
Sensor Combination	BMI270	BMI270 + BMM350	BMI270 + BMM350 + SPA06-003	BMM350
Sensor Type	Six-axis inertial sensor	Six-axis inertial + three-axis geomagnetic sensor	Six-axis inertial + three-axis geomagnetic + barometric pressure sensor	Three-axis geomagnetic sensor
Accelerometer	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Gyroscope	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Magnetometer	❌	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)
Barometric Pressure Sensor	❌	❌	SPA06-003	❌
I2C Address	0x68 / 0x69 (onboard switch)	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14 SPA06-003: 0x76	0x14
Main Functions	Acceleration, angular velocity, and temperature acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, and three-axis magnetic field acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, three-axis magnetic field, and pressure acquisition; on-chip motion detection	Three-axis magnetic field acquisition; heading output after calibration